

Algebra
Home Work 3

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A) Let G be a group and let $H, K \subset G$ be subgroups of G . Show that if $G = H \cup K$ then $H = G$ or $K = G$.

(*Hint:* Argue by contradiction. If $a \in H - K$ and $b \in K - H$, consider ab .)

Answer:

Given $H, K < G$ such that: $H \cup K = G$ (*).

Assume that $H \not\subset K$ and $K \not\subset H$ then $\exists h \in H \setminus K$ and $k \in K \setminus H$.

Then:

$$\begin{aligned} hk &\in G = H \cup K. \\ \Rightarrow \quad hk &\in H \quad \text{or} \quad hk \in K. \end{aligned}$$

If we assume $hk \in H$.

$$\begin{aligned} \Rightarrow \quad hk &= h' \quad \text{for some } h' \in H. \\ \Rightarrow \quad k &= h^{-1}h' \in H. \end{aligned}$$

Contradiction - which will also occur if we assume $hk \in K$

Hence, if (*) holds we must have $H < K$ or $K < H$.

$$\Rightarrow \quad H = G \quad \text{OR} \quad K = G$$

B) Let C_m and C_n be two finite cyclic groups of cardinality m and n respectively. Let $d = \gcd(m, n)$ be the greatest common divisor of m and n . We want to classify all morphisms of groups $f : C_m \rightarrow C_n$.

- (1) Show that, for every $x \in C_m$, $\text{ord}(f(x))$ is a divisor of d . (*Hint:* Show that this number is both a divisor of m and n .)
- (2) Let $C_d \subset C_n$ be the unique subgroup of cardinality d . Show that the image of f is contained in C_d . What happens if $d = 1$?
- (3) Fix a generator $g \in C_m$. Show that for every $y \in C_d$, there is a unique morphism of groups $f : C_m \rightarrow C_n$ such that $f(g) = y$.
- (4) What is the cardinality of the set of homomorphisms from C_m to C_n ?

Answer:

- (1) Let $x \in C_m \Rightarrow \text{ord}(x) \mid m$. But then $f(x)$ has $\text{ord}[f(x)] \mid m$ since:

$$\begin{aligned} [f(x)]^m &= f(x^m) = f(1) \\ &= 1 \end{aligned}$$

But $f(x) \in C_n \Rightarrow \text{ord}[f(x)] \mid n$. Hence $\text{ord}[f(x)] \mid n$ & $\text{ord}[f(x)] \mid m$

$$\Rightarrow \text{ord}[f(x)] \mid \gcd(m, n)$$

- (2) Let $K = f(C_m)$. Then every element of K has order that divides $d = \gcd(m, n)$.

$$\therefore K = f(C_m) \subset C_d \quad (\text{since } C_d \text{ is unique})$$

If $d = 1 \Rightarrow C_1 = \{1\}$ and C_1 only contains the identity element.

Hence

$$f(x) = 1 \quad \forall x \in C_m.$$

- (3) Let $\langle g \rangle = C_m$ and set $f(g) = y \in C_d$.

Then as C_m is cyclic every element is of the form g^k ($0 \leq k < m$). Hence

$$f(g^k) = [f(g)]^k = y^k$$

That is the map f is defined uniquely from $C_m \rightarrow C_d$ (and hence for $C_m \rightarrow C_n$) for every element in C_m .

- (4) The cardinality of the set of homomorphisms ($f : C_m \rightarrow C_n$) is defined by the number of different maps that map g (where $\langle g \rangle = C_m$) to an element of C_n that has order s , where $s \mid \gcd(m, n)$.

C) Let S be a set. A map $p : S \rightarrow S$ is called a *projector* if $p \circ p = p$. Let $G \subset \text{Map}(S, S)$ be a subset closed under composition (i.e., if $f, g \in G$, then $f \circ g \in G$) and such that $(G, \circ)$ is a group. (Note that we are not asking G to be a submonoid of $\text{Map}(S, S)$, i.e., G does not necessary contain Id_S .)

- (1) Show that G contains one and only one projector. Let p be the unique projector contained in G and let $T = p(S)$ be its image. Let

$$q = p|_{S-T} : S - T \rightarrow T$$

be the restriction of p to $S - T$.

- (2) Let $f \in G$. Show that $f(S) = T$ and that $f|_T : T \rightarrow T$ is a bijection. Show that $f|_{S-T} : S - T \rightarrow T$ is determined by $f|_T$ via the formula $f|_{S-T} = f|_T \circ q$. What is the inverse of f in G ?
- (3) Show that given a projector $p : S \rightarrow S$, there exists a largest subset $G(p) \subset \text{Map}(S, S)$, containing p , stable under composition, and such that $(G(p), \circ)$ is a group. Show that $G(p)$ is isomorphic to $(\text{Bij}(T, T), \circ)$, where $T = p(S)$.

Answer:

- (1) G is a group and $p \in G$. We need to show that p is the effective identity element for G and then its uniqueness is assured. We do this as follows:

$$\begin{aligned} p^{-1} \circ (p \circ p) &= p^{-1} \circ p \\ &= e \quad (\text{the identity for } G) \\ &= (p^{-1} \circ p) \circ p \\ &= 1 \circ p \\ &= p \end{aligned}$$

Hence, p acts as identity for G and is therefore unique.